

Mitchell Knoth

Cloud Software Engineer | Developer platforms, cloud infrastructure, and AI developer tools

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Cloud and developer-platform engineer building production systems for AI-assisted software delivery. Experience spans backend performance, cloud infrastructure, secure agent execution, and delivery governance.

EXPERIENCE - John Deere

AI-era software delivery | 2026 - present

- Wrote product requirements and much of the design record for a team-owned engineering evidence product now in production with steady, growing use.
- Delivered live-data views, a personal work timeline, and Redis caching with fail-open reads, tier-specific lifetimes, and freshness-versioned keys; main data endpoints became substantially faster.
- Separated CI validation from deployment and contributed dedicated acceptance checks for engineering views.
- Designed a shared knowledge repository with hybrid retrieval, a structural graph, a memory lifecycle, and CI lint checks; co-facilitate a guild and office hours for around 38 practitioners.
- Ran Codex and Claude Code in cloud sandboxes and added repository-credential proxying. Implemented a separate keyless Codex pilot; live verification and containment hardening remain in progress.
- Built a versioned skill-scanning trust layer with separate policy decisions and SARIF output; packaged reusable pre-PR reviews and feedback hooks.

Developer Experience | Late 2025 - early 2026

- Hardened the separation-of-duties platform with Datadog monitors, dashboards, and synthetics; added automatic incidents for pipeline failures and consolidated a legacy processor.
- Worked through CVE and Security Hub findings, migrated vulnerability-data integrations, reviewed organization governance, and introduced agent tooling into team repositories.

Enterprise cloud foundations | 2023 - 2025

- Led NAT Gateway consolidation, orchestrated a DNS migration, and automated certificate validation and renewal paths.
- Replaced manual account and VPC deletion with Step Functions and Lambda workflows; built DNS verification and cleanup and introduced reproducible Dev Container environments.

Digital equipment and external API platforms | 2019 - 2023

- Architected an external-facing dealer API with role-based authorization and audit logging; built a secure licensing-data ingestion prototype that helped make the case for an enterprise data-engineering investment.

TECHNICAL FOCUS

TypeScript, Python, Java; AWS, Terraform, Step Functions, Lambda; Redis; GitHub Actions; Datadog; identity and credential boundaries; agent harness and context engineering.

EDUCATION & SELECTED CREDENTIALS

B.S. Software Engineering, Iowa State University, 2020. Cum Laude; cyber security minor.

AWS Certified DevOps Engineer - Professional (May 2026)

AWS Certified Advanced Networking - Specialty (August 2025)